

#QUESTION 1 create variable 'name' and 'batch'.Next,assign 'Amrita Vishwa Vidyapeetham' and '2025' to be variables.Write the code for the following output:

```
In [1]: name='Amrita Vishwa Vidyapeetham'
batch='2025'
print("my college's name is",name,"and my batch is",batch)
```

my college's name is Amrita Vishwa Vidyapeetham and my batch is 2025

#QUESTION 2

```
In [3]: name='Shuptah'
age=17
print('my name is',name,'and my age is',age)
```

my name is Shuptah and my age is 17

#question3

```
In [4]: rupees='4500'
print("i don't have",rupees,"rupees in my wallet")
```

i don't have 4500 rupees in my wallet

```
In [20]: rupees="3000"
print('But i have "',rupees,'" rupees',sep="")
```

But i have "3000" rupees

#question4

```
In [27]: import math as m
a=1
b=4
c=3
r1=(-b+m.sqrt(b**2-4*a*c))/2*a
r2=(-b-m.sqrt(b**2-4*a*c))/2*a
r1,r2
```

Out[27]: (-1.0, -3.0)

#question5

```
In [1]: c=float(input('degree celcius'))
f=9/5*c+32
print('degree fareheit is',f)
```

degree celcius45
degree fareheit is 113.0

#question 6

```
In [7]: n=int(input('natural numbers:'))  
s=(n*(n+1))/2  
print('sum of n natural numbers',s)
```

```
natural numbers:10  
sum of n natural numbers 55.0
```

#question 7

```
In [8]: d=750  
tmin=2.5  
tsec=2.5*60  
s=d/tsec  
print('speed of the runner is',s)
```

```
speed of the runner is 5.0
```

#question 8 a

```
In [9]: a=int(input('first number'))  
b=int(input('second number'))  
print(a+b)
```

```
first number5  
second number6  
11
```

#question 8 b

```
In [10]: a=int(input('first number'))  
b=int(input('second number'))  
print(a*b)
```

```
first number7  
second number3  
21
```

#question 8 c

```
In [14]: s=int(input('side of the square'))  
A=s**2  
print('area of the square of side is',A)
```

```
side of the square8  
area of the square of side is 64
```

#question 8 d

```
In [15]: l=int(input('length of the rectangle'))
b=int(input('breadth of the rectangle'))
A=l*b
print('area of the rectangle is',A)
```

```
length of the rectangle15
breadth of the rectangle4
area of the rectangle is 60
```

#question 8 e

```
In [17]: import math
r=float(input('radius of the circle'))
A=math.pi*(r**2)
print('area of the circle is',A)
```

```
radius of the circle6
area of the circle is 113.09733552923255
```

#question 8 f

```
In [20]: a=float(input('first number'))
b=float(input('second number'))
c=float(input('third number'))
d=float(input('fourth number'))
A=(a+b+c+d)/4
print('average of four numbers is',A)
```

```
first number1
second number2
third number3
fourth number4
average of four numbers is 2.5
```

#question 8 g

```
In [22]: import math
r=float(input('radius of the arc'))
theta=float(input('angle of the arc'))
l=theta*r
L=theta*(math.pi/180)*r
print('length of arc in radians is',l,'\nlength of arc in degrees is',L)
```

```
radius of the arc6
angle of the arc60
length of arc in radians is 360.0
length of arc in degrees is 6.283185307179586
```

#question 8 h

```
In [24]: import math
r=float(input('radius of sphere'))
A=4*math.pi*(r**2)
V=(4/3)*math.pi*(r**3)
print('surface area of the sphere is',A,'\nvolume of the sphere is',V)
```

radius of sphere3
 surface area of the sphere is 113.09733552923255
 volume of the sphere is 113.09733552923254

#question 8 i

```
In [26]: import math
r=float(input('radius of the cylinder '))
h=float(input('height of the cylinder '))
A=(2*math.pi*r*h)+(2*math.pi*(r**2))
V=math.pi*(r**2)*h
print('surface area of the cylinder is',A,'\nvolume of the cylinder is',V)
```

radius of the cylinder4
 height of the cylinder5
 surface area of the cylinder is 226.1946710584651
 volume of the cylinder is 251.32741228718345

#question 9 a

```
In [6]: x=float(input('first variable = '))
y=float(input('second variable = '))
z=float(input('third variable = '))
expression=2.3*(3*x+(6*y+z**2)/3)
print('the value of given expression is',expression)
```

first variable = 1
 second variable = 2
 third variable = 3
 the value of given expression is 23.0

#question 9 b

```
In [8]: import math
a=float(input('first variable'))
b=float(input('second variable'))
c=float(input('third variable'))
expression=math.sqrt(5)*(((a**3)**(b**2)))+(2*a*c)-c)
print('the value of the given expression is',expression)
```

first variable2
 second variable3
 third variable4
 the value of the given expression is 300119990.4263926

#question 10 a

```
In [11]: import math
a=float(input('real part first number'))
b=float(input('imaginary part first number'))
c=float(input('real part second number'))
d=float(input('imaginary part second number'))
v1=complex(a,b)
v2=complex(c,d)
print('sum = ',v1+v2)
print('difference = ',v1-v2)
print('product = ',v1*v2)
print('division = ',v1/v2)
```

```
real part first number1
imaginary part first number2
real part second number3
imaginary part second number4
sum = (4+6j)
difference = (-2-2j)
product = (-5+10j)
division = (0.44+0.08j)
```

```
In [13]: print(v1.conjugate())
```

```
(1-2j)
```

```
In [14]: print(v1.real)
```

```
1.0
```

```
In [15]: print(v1.imag)
```

```
2.0
```

```
#question 11
```

```
In [17]: import math
a=float(input('first variable'))
b=float(input('second variable'))
r=math.sqrt(a**2+b**2)
theta=math.atan(b/a)
print('polar coordinates are (',r,',',theta,')')
```

```
first variable4
second variable5
polar coordinates are ( 6.4031242374328485 , 0.8960553845713439 )
```

```
#question 12
```

```
In [47]: import math
x=float(input('first variable'))
print(math.cos(math.sin(x**2)))
print(1/math.cos(1/math.tan(math.sqrt(math.e**x))))
print(math.log10(math.sin(math.pi/3)))
print(math.acos(0)+math.log(x,5))
```

```
first variable5
0.9912542848596704
-1.2716035555088556
-0.06246936830415
2.5707963267948966
```

In []: